

BEFORE / AFTER

One binary replaced three tools

BEFORE



AFTER



escpod absorbed the **POD5 CLI**, **Remora**, and the **leech charging backend** — one static Rust binary where there were three Python tools and a PyTorch stack. `pixi run setup` no longer installs torch at all.

The `ML→cl` retag step went with them: `escpod` writes its call directly onto the aligned records, so `dorado`'s modbase tags are never clobbered and never need restoring.

3 → 1
tools doing signal work

xN → x1
basecall passes per run

0
torch in the default install

12 GB
no longer duplicated

THE TWO NEW MODELS

What they actually do

LDX DEMUX

A CTC-CRF that **basecalls the barcode**

Reads the DNA barcode out of the raw adapter signal and matches the decode to references by **edit distance**, rather than classifying boundary-gated fingerprints as WarpDemuX does.

Those fingerprints barely separate — escpod's default segment-mean scores **0.03** discriminability, “no barcode signal”, and the whole ladder tops out near 0.49. A dead end, not a tuning problem.

The bundle is **self-describing** — its own references, geometry and pinned boundary detector — so no `--barcodes` or `--method` is ever passed.

16-plex
against 4–5 for WDX

12
min edit distance between refs

CHARGING

A recurrent net over the **offset axis**

Per-base features — signal mean plus a z-scored k-mer residual — over offsets `-8..+24` around the CCA–adapter junction, read as a **sequence** rather than a flat vector.

An MLP control with identical inputs but no offset axis lands below every architecture that has one: the gain belongs to **respecting that axis**, not to “neural beats trees”.

Labels are **library chemistry, not a model**: chemical ligation cannot attach to a deacylated tRNA, enzymatic needs a free 3'-OH.

It abstains where the common arm did not align — it scores 0.4993 there. Charging fractions over called reads alone run low.

MEASURED ON REAL DATA

What it bought

0.9906

charging AUROC, held-out flowcell

99.951%

demux agreement vs the prior run

395 KB

sidecar, replacing a full run copy

+0

reads unaccounted for in the split

Before the arm features were fixed, the charging model was **learning an artifact**: with arm availability derived from the aligner, and bwa stopping at the adduct, **20.9–25.3%** of charged-library reads lost the entire arm against **0.9%** of uncharged — so the model read that missing pattern as a class. Counting along the query instead removed it.

AND THREE THINGS THAT HAD BEEN FAILING QUIETLY

0

reads reported lost at charge-calling

...while **68 of 624** actually were.

100%

of final BAMs were invalid SAM

Every read cited an @RG that did not exist.

every

pipe ran without pipefail

A failed align exited 0 with an empty BAM.

All figures from 20,400 reads of the 2026-08-06 flowcell, all 16 LDX barcodes — not fixtures. The demux test path has no working fixture (issue #120), which is why. Measured pre-v6 basecall model; directional, not current.